



WELCOME





# Variables and Data Types

Recap

# Coding Practice Walkthrough | Part 1

- ✓ Coding Practice
  - Walkthrough
- ✓ Coding Question
  - Write
  - Run & Submit Code
- ✓ Other Features
  - AI Tutor
  - Save
  - Reset ...etc



Agenda for Today's Session

# Variables and Data Types



Variables

Data Types

Constants

# Variables

# What is a Variable?



- ✓ A **variable** is essentially a container used to **store data**
- ✓ It can be **accessed** and **modified** during program execution

# What is a Variable?



- ✓ Python and JavaScript are **dynamically typed** languages
- ✓ Type of variable is **decided at runtime**

# What is a Variable?

## Example in JavaScript

Javascript

```
let age = 10;  
let sport = "Cricket";
```

## Example in Python

Python

```
age = 10  
sport = "Cricket"
```



# Defining a Variable in JavaScript



- ✓ **Variables** are created using keywords like **let**, **const**, or **var**
- ✓ Their type is determined by the assigned value (e.g., **number**, **string**)
- ✓ **Statements** may end with **semicolons**, though they are **optional**

# Defining a Variable in Python



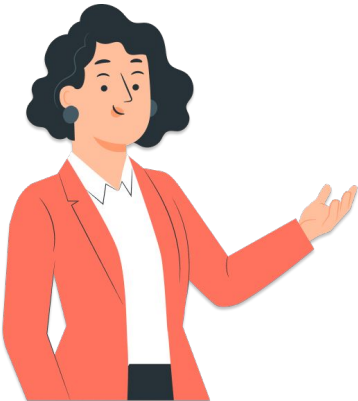
- ✓ In **Python**, **variables** can be **created directly** without any **keywords**
- ✓ A variable is created **automatically** when you assign a value to it

# Example

Python

```
age = 10  
sport = "Cricket"
```

The moment you assign **age** a **value (age = 10)**, the **variable age** is created and the **value 10** is **stored** in it



# Variable Naming Rules & Conventions



- ✓ **Must start with** a letter or an underscore ( \_ )
- ✓ **Cannot start with** a number
- ✓ **Case-Sensitive:** Variables like **age** and **Age** are considered different
- ✓ **No Keywords:** You can't use Python keywords like **print**, **if**, or **True** as variable names

# Variable Naming Rules & Conventions

## Key Difference



### JavaScript

Variable names can start  
with a **dollar sign (\$)**



### Python

A **variable** cannot start  
with a **dollar sign (\$)**

# Variable Naming Rules & Conventions

## Best Practices for Naming Variables



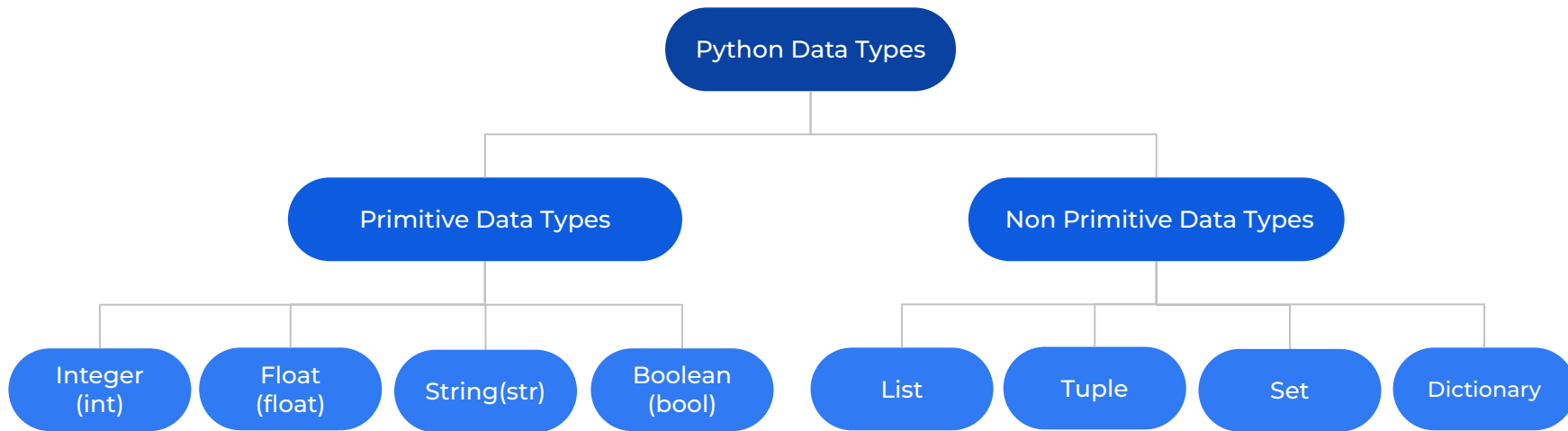
**Snake case** is preferred (e.g., `user_age`)



**Camel case** is used (e.g., `userAge`)

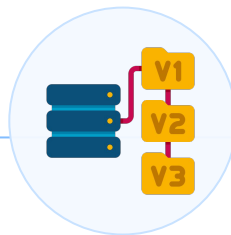
# Data Types

# Data Type Classification





# Primitive Data Types

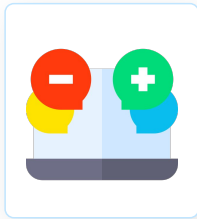


- ✓ Primitive types are basic data types like **numbers**, **strings**, and **booleans**
- ✓ They hold single values and form the **building blocks** of **data** in **programming**



# Integer

Integer (int): Whole numbers



Represents **positive** , **negative** numbers and **Zero**  
**without** a **fractional** part

# Integer

Python

```
age = 42  
print(age)
```

Javascript

```
let age = 42;  
console.log(age);
```

Output

42



Both in **Python** and **JavaScript**, **integers** represent **whole numbers**.

# Float

Float (float): decimal points



Represents **numbers** with a **fractional** part (e.g., **3.14**, **2.5**)

# Float

Python

```
height = 5.8  
print(height)
```

Javascript

```
let height = 5.8;  
console.log(height);
```

Output

5.8



In **Python** and **JavaScript**, **floats** represent **decimal numbers** for precise calculations

# String

String (str): A sequence of characters



Represents **text** enclosed within **quotes** (single or double)

## Primitive Data Types

# String

Python

```
greeting = "Hello, world!"  
print(greeting)
```

Javascript

```
let greeting = "Hello, world!";  
console.log(greeting);
```

Output

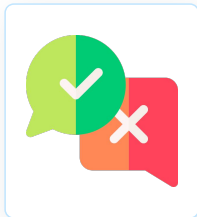
```
Hello, world!
```



Both **Python** and **JavaScript** use **strings** to represent **text**,  
**enclosed** in either **single** or **double** quotation marks

# Boolean

Boolean (bool): Represents True or False



Represents a **binary state** (True/False)



## Primitive Data Types

# Boolean

Python

```
is_active = True  
print(is_active)
```

Output

True

Javascript

```
let is_active = true;  
console.log(is_active);
```

Output

True



In both **Python** and **JavaScript**, **booleans** represent **binary values**

# None

**None (NoneType): Represents absence of a value**



**None** is **Python's equivalent** of **null**, representing the **absence of a value**

## Primitive Data Types

# None

Python

```
value = None  
print(value)
```

Output

None

Javascript

```
let value = null;  
console.log(value);
```

Output

Null



While **None** is used in **Python** to signify **no value** or an **empty state**, **null** is used in **JavaScript** for **similar purpose**

# Non Primitive Data Types



We will learn **Non-Primitive Data Types** in upcoming sessions

# Quiz Time!



# type() Function in Python



- ✓ `type()` is a **built-in function** in **Python** that allows you to **determine** the **type** of a **variable**
- ✓ It is used to **check** what **data type** a **variable** holds,
- ✓ Helpful for **debugging** or for **ensuring** that the variable is of the **expected type**

# type() Function in Python

Python

```
age = 42
print(type(age))

name = "Alice"
print(type(name))
```

Output

```
<class 'int'>
<class 'str'>
```

Here, the `type()` function is used to print the type of `age` and `name`, which are an `int` and a `str` respectively

# None in Python



- ✓ **None** is a **keyword** in **Python** that represents the **absence of a value** or a **null state**
- ✓ You often encounter **None** when a **variable** is **declared** but **not initialized**



# None

Python

```
value = None  
print(type(value))
```

Output

```
<class 'NoneType'>
```



# None

## Variable Value

Assigned None

## Type

NoneType (confirmed by `type()`)

## Meaning

Represents the absence of a value

## Usage

Useful for initializing variables without meaningful data

# Python vs JavaScript

## Comparison of Data Types

| Feature       | Python                                                                                    | JavaScript                                                        |
|---------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Number Types  | Two types: <b>int</b> (for whole numbers) and <b>float</b> (for decimals)                 | One type: <b>number</b> handles both ints and floats              |
| Booleans      | Written as <b>True</b> and <b>False</b> (capitalized)                                     | Written as <b>true</b> and <b>false</b> (lowercase)               |
| Undefined     | Python uses <b>None</b> to represent the <b>absence</b> of a <b>value</b> or a null value | Variables that are declared but not assigned are <b>undefined</b> |
| Type Checking | Uses <b>type()</b> function                                                               | Uses <b>typeof</b> operator                                       |

# Constants

# Constants in JavaScript



JavaScript provides a **keyword** called **const** to **declare constants**



Used when the **value** should **remain fixed**  
**throughout** the **program**

# Constants in Python



- ✓ Python does **NOT** have a **const keyword**
- ✓ Python does not provide any **built-in feature** to **create true constants**
- ✓ Developers follow a **naming convention** to **indicate constants**

# Convention Used in Python

Constants are written in **UPPERCASE** to indicate they should **not be changed**

## Example

Python

```
PI = 3.14
```

# Convention Used in Python



Even though reassignment is **possible**, it is **not recommended**

```
PI = 3.14159
```

Python



Allowed, but  
discouraged



Constants

# Constants in JavaScript

## Example

Javascript

```
const Pi = 3.14;
```

Output

```
TypeError{"Assignment to constant  
variable."}
```

✗ Reassigning is not allowed

Javascript

```
Pi = 3.14159;
```



# Constants

## Key Difference



### JavaScript

constants are supported  
using **const**



### Python

constants are **by  
convention only**, not  
enforced by the language

# Quiz Time!



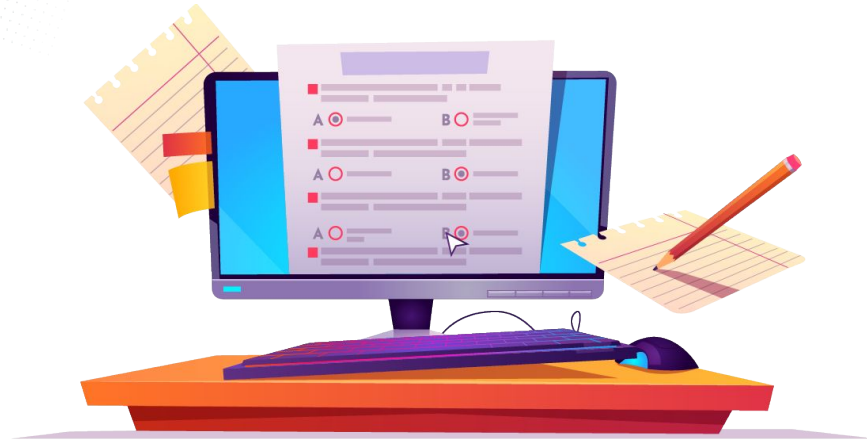
Variables and Data Types

# Key Takeaways

- ✓ Variables
- ✓ Data Types
- ✓ Constants



# Practice



Next Session

# Sequence of Instructions



THANK YOU





ALL THE BEST

